

December 2025 Linux+ Class Review

Command Help Tools

man: displays detailed information about the commands including syntax, options, examples, and descriptions

help: shows brief usage/syntax for the commands

Fundamental Commands

cd (change directory): changes the folder we are currently working out of

pwd (print working directory): outputs the current file path we are working out of

ls (list): displays the content of a directory/folder

cat: outputs the contents of a file

grep: searches for specific strings within a file

mkdir: creates a folder

touch: creates an empty file

nano: opens the Nano text editor to modify text files

Debian vs RHEL

Debian relies on community driven development while RHEL has enterprise backing.

systemctl VS systemd

systemctl: sends instructions to the programs (e.g., disabling SSH)

systemd: the background programs themselves (e.g., SSH)

chmod vs umask

chmod: chmod is the command used to add permissions for existing files.

umask: umask is the command used to set the default permissions for new files that are created.

umask subtracts from the permission defaults so it takes away access, which is the opposite of chmod.

Permission Values

4: read (open the file)

2: write (modify the file)

1: execute (allows the file to be run as a script or programs)

Example

chmod 750

- 7: Sets read, write, and execute permissions for the owner of the file
- 5: sets read and execute permissions for group members.

- 0: denies access permissions for everyone else

umask 021

- 0: grants read, write, and execute permissions for the file owner.
- 2: removes write permission for the group, and sets the permissions as read and execute
- 1: removes execute permission for everyone else, and sets the permissions as read and write.

Multifactor Authentication

Authentication: verifying who you are (e.g., checking your ID)

Authorization: grants or denies access to organization resources

Accounting: user activities are logged

Public Key Infrastructure (PKI)

Public Keys: publicly shared keys used to encrypt data or verify signatures

Private Keys: secret keys held only by the owner, used to decrypt data or create signatures

Self-signed certificates: servers used their own generated certificates

Digital Signatures: proves that a digital message / action originated from a specific individual and has not been altered

Wildcard Certificates: a certificate used with multiple subdomains of a domain (e.g., *.infotec.com)

Hashing: one-way function that transforms data into a fixed-size value to ensure integrity and detect changes

Certificate Authorities: trusted third parties that issue and validate digital certificates for secure identity verification

Storage

PVG (Physical Volume Group): the storage hardware

VG (Volume Group): a group of storage that is created by physical volumes

LV (Logical Volume): the sections of the hard drive

RAID

Mirroring: replicates a full backup of the data

Parity: a calculated checksum is stored with data blocks for recovery after a drive failure

Striping: splits data into blocks and writes them across multiple disks

0: Striping, minimum number of drives = 2

1: Mirroring, minimum number of drives = 2

5: Striping + Parity, minimum number of drives = 3

10: Striping + Mirroring, minimum number of drives = 4

Software Management: APT is used with Debian and DNF is used with RHEL Linux versions

Debian/Ubuntu (APT)

sudo apt install nmap (installs nmap)

sudo apt remove nmap (uninstalls nmap, keeps the configuration file)

sudo apt purge nmap (uninstalls nmap and deletes the configuration file)

sudo apt update && sudo apt upgrade (refreshes package lists and upgrades installed packages)

Fedora/RHEL (DNF)

sudo dnf install nmap (installs nmap)

sudo dnf remove nmap (uninstalls, keeps config)

sudo dnf autoremove nmap (uninstalls fully, cleans dependencies)

Network Troubleshooting

ping: test the connection between two devices (it's like saying hi to your neighbor to see if they're home)

ifconfig: allows you to view the IP address of your device (gives a similar output to the Windows ipconfig command.)

tracert: outputs the path the packet took to arrive to its destination.

nmap: allows you to scan your network for open ports, see which devices are connected, and the operating system versions being used by the systems.

nmcli: view your network configurations and make changes

ip addr show: shows the network

mtr (my traceroute): shows network routing information in real-time

nc (netcat): allows you listen to live network traffic (Wireshark is the version with the Graphical User Interface)

dig: provides detailed DNS information about a site and its DNS server

Performance Troubleshooting

ps -a: view all the running process

free: view the memory that is currently use.

top: view a live list of processes that are running on the system.

df (disk free): shows the total storage, used space, and available storage

du (disk usage): shows the space that specific files and directories are taking up

iostat: monitors CPU usage an I/O statistics

Boot Process in Order

BIOS/UEFI: Firmware on motherboards that runs first at power-on, performs hardware checks (POST), and loads the bootloader from disk.

Bootloader: Small program that loads the kernel into memory and hands off control during system startup (e.g., GRUB).

- **GRUB:** Old bootloader that breaks if drive order changes and lacks modern features like scripting
- **GRUB2:** Newer version with stable naming, UEFI support, themes, and custom scripting capabilities

Kernel: OS core that controls hardware, runs programs, and manages resources after boot.

Init: The root process PID (Process ID) 1 starts services, runlevel, and user login

Backup Types

Full: copies all data

Differential: copies only the files that have changed since the last full backup

Incremental: only copies the changes to the data since the last full or incremental backup

Scheduling Automated Tasks

Cron: used to schedule jobs that repeat (e.g., weekly backups)

At: used to schedule a one time job (e.g., one-time system restart scheduled at 2AM)

Scripting

while: repeats an action as long as the specified parameters remain true

for: executes an instruction until it is run on all specified values in a list

until: performs an action until a specified parameter becomes true

if: executes instructions based on if a parameter is true or false

switch/case: used for configuring menu options

Cloud and Virtualization

Infrastructure as a Service: cloud model providing virtualized computing resources (servers, storage, networking) on demand.

Platform as a Service: Cloud service providing managed servers, databases, and dev tools so you focus on coding apps without handling OS or hardware.

Software as a Service: cloud model delivering fully managed applications over the internet, accessible via a web browser without local installation.

Type 1 Hypervisor: bare-metal software running directly on hardware to manage VMs with high performance and security (e.g., ESXi, Hyper-V).

Type 2 Hypervisor: hosted software running on top of a host OS to create VMs, easier for testing but with OS overhead (e.g., VirtualBox).

Virtual Machines: fully emulated hardware environments running isolated guest operating systems and applications on a hypervisor.

Containers: lightweight, OS-level virtualization packaging apps with dependencies for fast deployment without full OS emulation.

Links for Additional Study Resources

<https://www.stationx.net/bash-cheat-sheet/>

<https://www.youtube.com/watch?v=o-83E6levzM>

<https://www.booleanworld.com/raid-levels-explained/>

[https://youtube.com/playlist?list=PLD2g03RR87vgVRUUrnFAP8HXRIEixyusc&si=Q26DTgB5AppKT
w0f](https://youtube.com/playlist?list=PLD2g03RR87vgVRUUrnFAP8HXRIEixyusc&si=Q26DTgB5AppKT
w0f)